FDA FAERS ANALYTICS · AI + BIG DATA

DETECT HIDDEN DRUG SAFETY RISKS FASTER WITH AI

From millions of messy adverse event reports to actionable safety insights — in seconds.

TEAM 11

BUSINESS ISSUE

Large Data Volume
~1 million new FAERS reports filed every year

Inconsistent Data
Unstructured, free-text user entry with no standardization

Poor Usability
Current dashboard lacks granularity and is hard to navigate

Clinical Terminology
Medical jargon limits access for non-specialist users

OUR SOLUTION

CENTRALIZED STORAGE
- FAERS public quarterly data extracts
- Scalable cloud data processing
- Databricks & Spark for dedup and aggregation

AI-POWERED MODELING
- LLM normalization of messy drug names
- Semantic clustering of related adverse event terms
- AI-generated summaries of flagged safety signals

REAL-TIME INSIGHTS
- Streamlit interactive dashboard
- Drill-down by drug, event, and trend
- Reviewer-facing explanations for flagged cases

DATA PIPELINE

Ingest — Quarterly FAERS files → **Clean** — Deduplicate in Spark → **Standardize** — Drug & reaction terms → **Identify** — Key drugs, manufacturers & reactions → **Surface** — Streamlit dashboard

CORE CAPABILITIES

SMART SEARCH
Semantic mapping — "headache" finds cephalalgia, migraine, and related terms automatically

RISK DETECTION
Identify unexpected drug-event relationships and surface emerging safety signals

SIGNAL SUMMARIZATION
AI-generated plain-language summaries of adverse event clusters for non-clinical users

INTERACTIVE UI
Drill into drugs, events, and trends — filter and explore in real time

BUSINESS VALUE & TECHNICAL HIGHLIGHTS

Turning FAERS complexity into competitive advantage for pharma, analysts, and healthcare systems

BUSINESS VALUE BY AUDIENCE

PHARMA COMPANIES

Competitive Safety Intelligence

- Monitor competitor drug safety issues
- Improve risk management decisions
- Proactively address adverse trends

ANALYSTS & REGULATORS

Faster Public Health Risk ID

- Rapid identification of public health risks
- Dramatically reduce manual review effort
- AI-assisted signal summarization

HEALTHCARE SYSTEMS

Adverse Event Trend Awareness

- Better awareness of adverse event patterns
- Support clinical decision-making
- Accessible to non-clinical staff

TECHNICAL HIGHLIGHTS — 3V FRAMEWORK

VOLUME

MASSIVE SCALE, HANDLED EFFICIENTLY

- ✓ **In-Database Power**
  Spark processes millions of FAERS records without export
- ✓ **Efficient Big-Data Handling**
  Databricks dedup and aggregation at scale

VELOCITY

REAL-TIME PROCESSING & INSIGHT

- ✓ **Blazing Bulk Loads**
  Quarterly FAERS ingestion in minutes, not days
- ✓ **Elastic Compute, Zero Tuning**
  Auto-scaling clusters via Databricks

VALUE

INSIGHT WITHOUT HEAVY IT LIFT

- ✓ **Low-Code, High-Impact**
  Streamlit UI requires no engineering to operate
- ✓ **Cloud-Smart Economics**
  Pay-per-use Databricks — no infrastructure lock-in

GENERATIVE & AGENTIC AI LAYER

- Normalize messy free-text drug names to standardized identifiers
- Summarize clusters of related adverse events into readable safety signals
- Generate reviewer-facing explanations of why a case or signal was flagged

INDUSTRY APPLICATIONS — WHEREVER DATA DRIVES DECISIONS

🧪 PHARMA & BIOTECH

Post-market surveillance and competitive safety monitoring
e.g. Track competitor adverse event trends in real time

⚖️ REGULATORY AGENCIES

Accelerate signal review and prioritize high-risk cases
e.g. FDA analysts surface emerging risks in hours

🏥 HEALTHCARE SYSTEMS

Equip clinical teams with adverse event trend awareness
e.g. Hospital pharmacists monitor drug safety dashboards

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SCAN TO SEE THE DASHBOARD